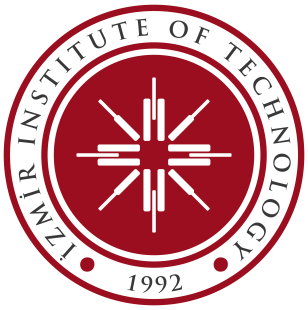


İzmir Institute of Technology

Faculty of Engineering

Department of Electrical and Electronics Engineering

EE 315 Electronics Laboratory Prelab Sheet

Experiment 2

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1 LTspice Simulation

1.1 Full-Wave Bridge Rectifier

The purpose of a full-wave bridge rectifier circuit is to convert an AC voltage to a DC voltage. As the bridge rectifier rectifies both half-cycles of the input signal into a single polarity, the frequency of the output voltage is doubled relative to the input frequency. Since two diodes conduct in series during each half-cycle, the circuit has a voltage drop equal to the sum of the forward voltage drops of the two conducting diodes. To prevent severe voltage sag, a filter capacitance is needed. An example circuit is given in Figure 1.

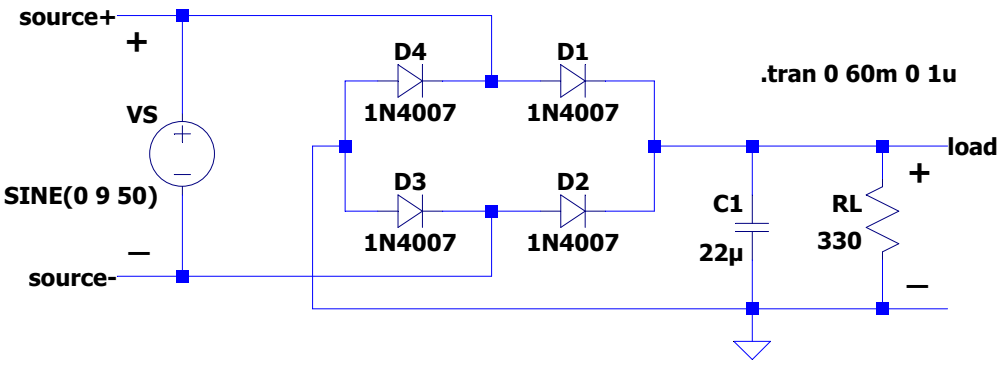


Figure 1: Schematic of full-wave bridge rectifier circuit

The input and load voltage of the example circuit is given in Figure 2.

R_L	$V_{L,\text{peak}}$	f_L	**Number of conducting diodes**
330 Ω	7.59 V	100 Hz	2

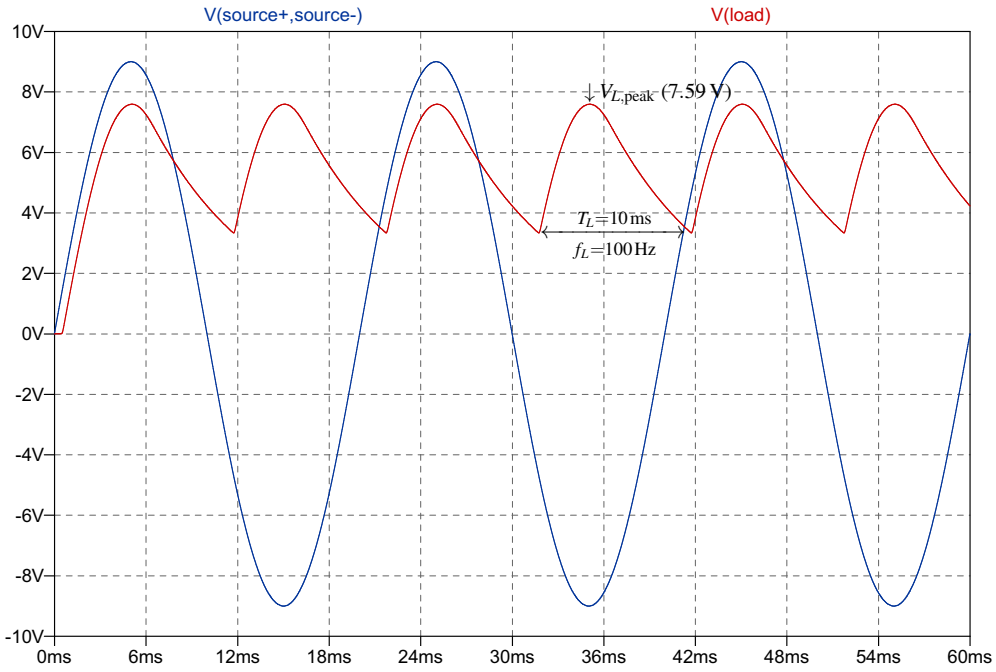


Figure 2: Input and output voltage waveforms of the full-wave bridge rectifier circuit ($R_L = 330\,\Omega$)

1.2 Zener Diode Regulator

In full-wave bridge rectifier circuits, a key limitation is that the peak output voltage ($V_{L,\text{peak}}$) and the peak-to-peak ripple voltage ($V_{rpl,PP}$) are affected by load conditions. As the load current increases, the voltage drops across the bridge diodes and series components increase, causing the output voltage to sag. To address this limitation and clamp the peak output voltage to a desired, more stable level, a Zener diode regulator can be implemented. However, since the excess voltage is entirely dropped across the series resistor (R_S), this circuit suffers from poor thermal and power efficiency, especially in high-current applications. Despite this inefficiency, it remains a low-cost practical solution for low-current voltage references and basic low-power circuits. An example circuit is given in Figure 3.

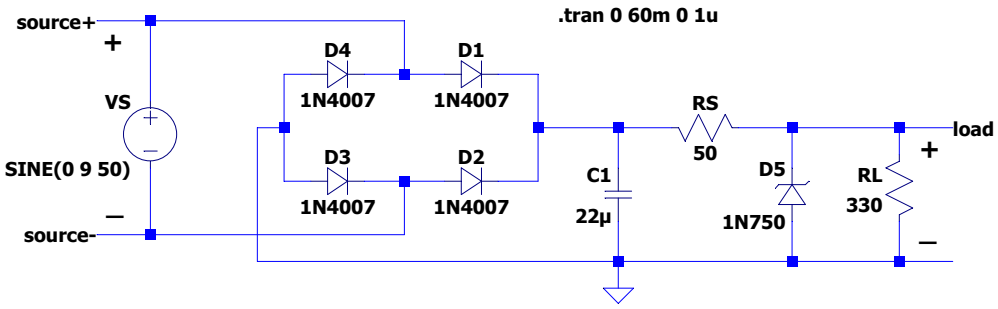


Figure 3: Schematic of Zener diode regulator circuit

The input and load voltage of the example circuit is given in Figure 4.

Breakdown Voltage (V_Z)	$V_{o,DC}$	I_Z
4.7 V	4.13 V	−13.15 mA

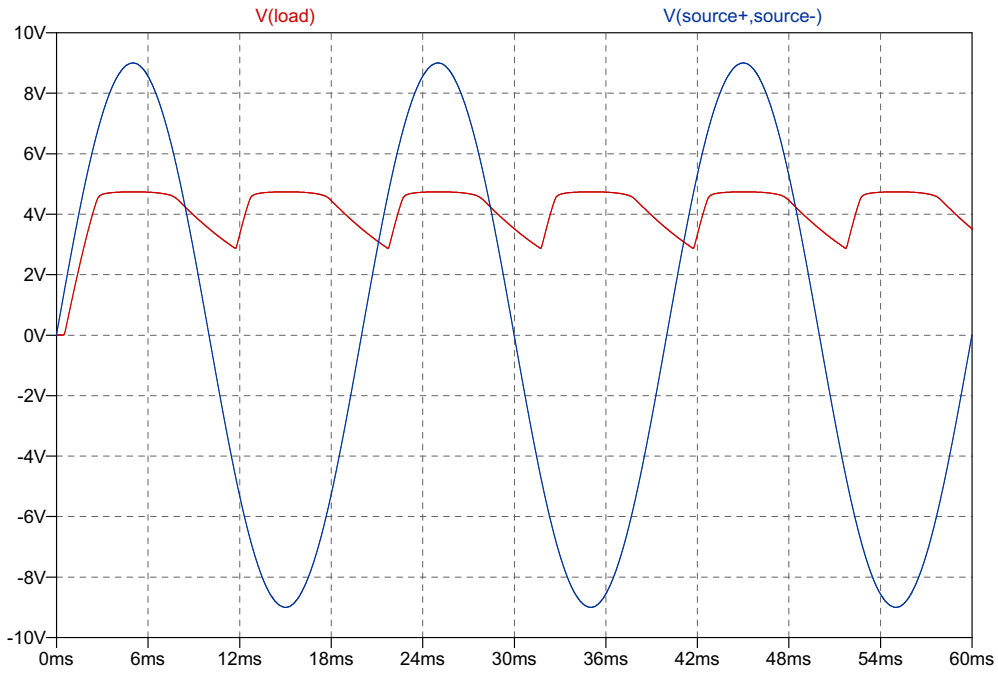


Figure 4: Input and output voltage waveforms of the Zener diode regulator circuit ($R_L = 330\,\Omega$)